

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

AI in Population Health Management (PE-4) [CSE_4015]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the Healthcare Reference Model to design a workflow for a new patient admission, showing how general patient data, Process steps and validation steps interact to improve efficiency. (4)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A large healthcare facility introduces an AI-powered nursing management system designed to support nurses in planning and delivering patient care. The system uses predictive analytics to identify patients at risk of complications, NLP tools to interpret nursing notes, and clinical decision support algorithms to suggest personalized care interventions. It also enables real-time communication and coordination between nursing teams, physicians, and other healthcare staff. Analyse how AI can be applied in nursing plans to improve patient outcomes and strengthen care coordination. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) List the components of the Healthcare Reference Model. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) A multispecialty hospital integrates AI-powered diagnostic tools-such as image recognition systems for radiology and predictive analytics for pathology-to assist clinicians in detecting diseases more accurately and efficiently. Over time, these tools begin to influence clinical decision-making, with doctors increasingly relying on AI-generated insights for diagnosis and treatment planning. However, some patients express concerns about transparency, data privacy, and overdependence on technology, raising questions about trust and accountability in AI-assisted care. Explain the impact of AI-powered diagnostic tools on clinical decision-making and patient trust. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- B) An advanced cancer research hospital is exploring ways to personalize treatment plans for oncology patients. To achieve this, it implements an AI-based cognitive computing system (such as IBM Watson for Oncology) that analyzes clinical records, genetic data, pathology reports, and medical literature. The system recommends evidence-based treatment pathways tailored to each patient's cancer type, stage, and response to therapy. Clinicians use these insights to design individualized care plans and monitor treatment outcomes in real time. Apply AI-based cognitive computing techniques to design an effective treatment pathway for oncology patients. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Explain how DataDerm supports quality assessment in dermatology. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) An emergency room uses an AI-based CT triage system to detect acute brain haemorrhages. (4)
- A) Researchers find that minor, invisible pixel changes cause AI to misclassify severe haemorrhages as normal. Analyse how technological limitations of AI affect its performance in critical healthcare environments.
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) An AI system is trained to detect skin cancer using a dataset consisting mainly of images of light-skinned patients. When deployed in a multicultural hospital, the system consistently underdiagnoses melanoma in darker-skinned individuals. Apply the concept of AI bias to demonstrate how judgmental datasets can influence healthcare outcomes. (4)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Outline the adversarial examples in AI. Give one potential implication of AI being fooled by adversarial inputs. (2)
- Course Outcome: "CO3" / Cognitive Level: "Understand"
- 4) Analyze the differences between the three types of process mining in the healthcare context. (4)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Apply the concept of process classification to organize different healthcare processes at various levels of care. (4)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- C) Explain the difference between Model Level, Instance Level, and Event Level in process mining. (2)
- Course Outcome: "CO4" / Cognitive Level: "Understand"
- 5) During a surge in COVID-19 cases, a hospital's radiology department is overwhelmed with patients requiring rapid diagnosis. To support radiologists, the hospital deploys AI-based image processing systems that analyze chest X-rays to differentiate between COVID-19 pneumonia and other respiratory infections. Two different AI models - a CNN-based system and a transfer learning-based model (such as ResNet or VGGNet) are being evaluated for accuracy and speed. Compare AI-based image processing methods to explain how chest X-rays can be used to detect COVID-19 pneumonia. (4)
- A) Course Outcome: "CO5" / Cognitive Level: "Application"
- B) Apply the concept of AI in the drug discovery lifecycle to demonstrate how machine learning accelerates target identification and validation. (4)
- Course Outcome: "CO5" / Cognitive Level: "Evaluate"
- C) Summarize Explainable AI (XAI) in drug discovery with benefit of using PADME and DeepAffinity. (2)
- Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4015:AI in Population Health Management (PE-4)

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4015:AI in Population Health Management (PE-4)

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	10(20%)
3	Application	5	20(40%)
4	Analysis	3	12(24%)
5	Create	2	8(16%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)